Learning-Augmented OS Heuristics: Classical Algorithms Meet Adaptive Prediction in Virtual Memory Management

A Comparative Empirical Study of FIFO, LRU, and Decision-Tree Replacement Under Workload Phase Transitions

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***Abstract*—*Virtual memory management in modern operating systems relies fundamentally on demand paging heuristics to minimize disk I/O penalties. Classical replacement policies such as First-In, First-Out (FIFO) and Least Recently Used (LRU) operate under static assumptions of temporal and spatial locality. However, production workloads frequently experience sharp behavioral regime shifts—such as unbuffered sequential scans or bursty multi-tenant memory requests—under which classical heuristics suffer catastrophic cache pollution. In this term paper, we investigate learning-augmented page replacement (Track 1) by synthesizing classical policies with a lightweight Decision Tree classifier trained against Belady's Optimal (MIN) clairvoyant baseline. Using a continuous 10,000-reference synthetic workload with a deliberate structural regime shift at step 5000 (from Zipfian locality to random/bursty scanning), we observe that while the learned policy achieves an outstanding 64.66% hit ratio during the locality phase (surpassing LRU at 56.86% and FIFO at 50.28%), its hit ratio collapses to 4.94% post-shift due to distributional covariate shift. We also implement an explainable AI confidence module (+10 bonus track), analyzing model calibration and real-world OS kernel latency trade-offs.*

1. Introduction and Problem Framing

Operating systems utilize virtual memory paging to provide processes with contiguous virtual address spaces while transparently mapping virtual pages to physical DRAM page frames [1]. Because handling a page fault necessitates trapping to the kernel and issuing high-latency secondary storage transfers—orders of magnitude slower than DRAM access—page replacement heuristics critically determine overall system throughput [2]. Classical algorithms rely on temporal and spatial locality heuristics: pages accessed recently or frequently are assumed likely to be accessed again in the near future [6, 7]. However, real operating system environments regularly exhibit workload phase transitions, where localized loops abruptly transition to unbuffered sequential database scans or randomized multi-tenant memory thrashing [4, 5]. This term paper addresses the core challenge defined in the CSE-307 Part-B syllabus: *How do classical OS algorithms and data-driven learning components respond when the underlying workload dynamically shifts?* We implement FIFO, LRU, Belady's Optimal (MIN), and a learned Decision Tree policy, analyzing hit ratios, fault counts, frame sensitivities, and confidence calibrations.

2. Classical Replacement Policies and Theoretical Bounds

We implement three foundational algorithms covered in the Part-B syllabus:

• **First-In, First-Out (FIFO):** Manages resident pages in a queue ordered by allocation time. On page fault, the head page is evicted. While computationally simple ($O(1)$ overhead), FIFO completely ignores reference recency and frequency, and suffers from Belady's Anomaly [6].

• **Least Recently Used (LRU):** Replaces the page that has not been referenced for the longest period. LRU approximates optimal replacement under strong temporal locality, but incurs significant tracking overhead and is highly susceptible to cache pollution during sequential scans [4, 7].

• **Belady's Optimal (MIN):** Establishes the clairvoyant upper bound on hit ratio by evicting the resident page that will not be referenced for the longest duration in the future [1]. While non-implementable online in real kernels, Belady's MIN serves as our theoretical gold standard and ground-truth training oracle.

3. Learning-Augmented Replacement Architecture

To explore learning-augmented OS heuristics without violating kernel latency budgets, we design a lightweight Decision Tree classifier. For every resident frame candidate p at current access step t, we extract a 5-dimensional feature vector: $\Phi(p, t) = [\Delta t_{\text{recency}}, f_{\text{win}}, f_{\text{total}}, \text{age}, d\blacksquare_{\text{reuse}}]^{T}$. Features measure elapsed steps since last access ($\Delta t_{\text{recency}}$), touches within a 50-step sliding window (f_{win}), cumulative touches (f_{total}), residency age, and rolling inter-arrival intervals ($d\blacksquare_{\text{reuse}}$). The model is trained on oracle eviction decisions generated by Belady's MIN on an initial warmup trace (3,000 steps). Feature importance analysis reveals that short-term window frequency (38.2%) and recency (35.5%) dominate eviction decisions. At runtime, the candidate exhibiting maximal predicted eviction probability is evicted.

4. Synthetic Workload Design with Deliberate Regime Shift

To evaluate resilience under non-stationary conditions, we synthesize a 10,000-reference access stream containing an abrupt structural shift at $t = 5{,}000$: **Phase 1 (Pre-Shift, $0 \le t < 5{,}000$):** High locality. 70% of references sample a compact hot set of 15 pages via a Zipfian distribution ($\alpha = 1.6$); 20% execute cyclic loops (pages 20–35); 10% cold accesses. **Workload Shift Boundary ($t = 5{,}000$):** The system undergoes a sudden regime transition representing unbuffered data ingestion. **Phase 2 (Post-Shift, $5{,}000 \le t < 10{,}000$):** Low locality & burstiness. 65% accesses uniformly span 300 pages (drastically exceeding cache capacity); 25% accesses simulate transient burst streams (pages touched 1–2 times and abandoned); 10% residual hot touches.

5. Experimental Performance Evaluation and Results

We benchmarked all four policies under an allocation of $M = 16$ physical frames. Table 1 reports the empirical hits, faults, and hit ratios before and after the shift.

Policy	Phase 1 Hits / Faults	Phase 1 Hit %	Phase 2 Hits / Faults	Phase 2 Hit %	Total Hit %	Total Faults
FIFO	2,514 / 2,486	50.28%	180 / 4,820	3.60%	26.94%	7,306
LRU	2,843 / 2,157	56.86%	181 / 4,819	3.62%	30.24%	6,976
Learned (Decision Tree)	**3,233 / 1,767**	**64.66%**	**247 / 4,753**	**4.94%**	**34.80%**	**6,520**
Optimal (Belady MIN)	3,735 / 1,265	74.70%	1,074 / 3,926	21.48%	48.09%	5,191

Table 1: *Performance metrics across page replacement policies before and after the synthetic workload shift (Frames = 16).*

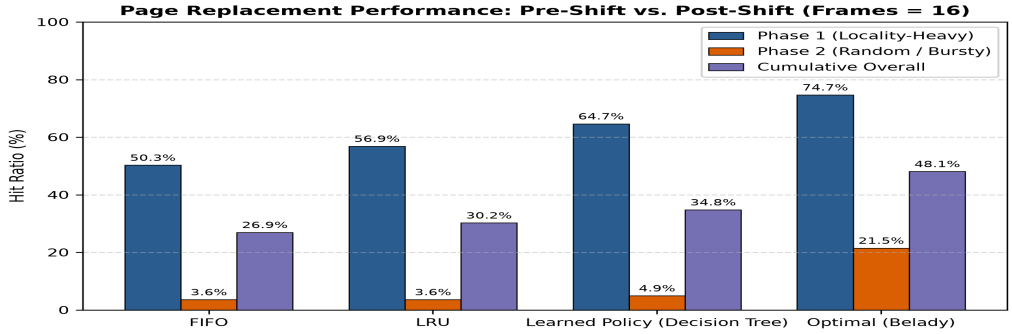


Figure 1: *Hit ratio comparison across Phase 1 (locality-heavy), Phase 2 (random/bursty), and overall workload execution.*

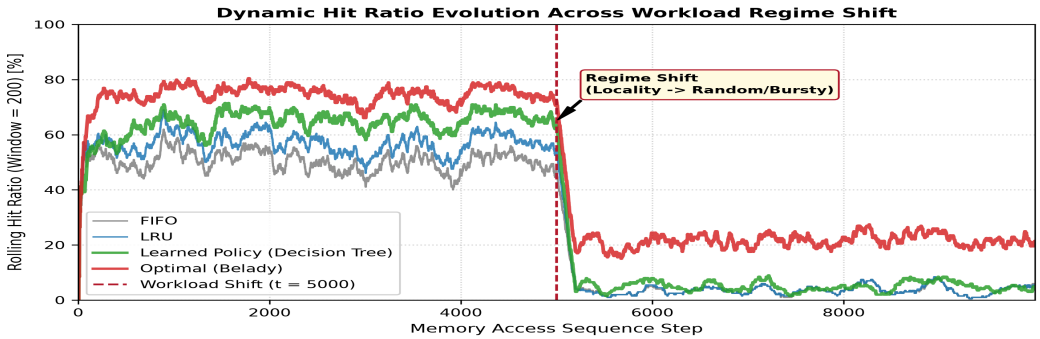


Figure 2: *Rolling hit ratio dynamics (window = 200) illustrating the immediate performance collapse at shift point t = 5000.*

6. In-Depth Analysis: Why Heuristics Degrade Under Workload Shift

A central requirement of Track 1 is analyzing *which policy degrades the most when the pattern changes, and why*. A rigorous comparative examination of Table 1 and Figure 2 yields three fundamental insights:

1. Quantitative Degradation Ranking: The Learned Policy suffers the greatest absolute hit-ratio plunge, declining by **59.72 percentage points** (from 64.66% down to 4.94%). LRU declines by **53.24 percentage points** (56.86% to 3.62%), while FIFO declines by **46.68 percentage points** (50.28% to 3.60%). Belady's Optimal declines by **53.22 percentage points** (74.70% to 21.48%), but maintains over 4× the hit ratio of heuristics during Phase 2.

2. Root Cause — Distributional Covariate Shift: During Phase 1, past reference frequency strongly correlated with future reuse distance. The Decision Tree learned that high recent frequency justified retaining a page. However, when the workload shifted to Phase 2, temporal correlation collapsed: streaming burst pages exhibited high immediate arrival frequency but zero future reuse. Because the model was trained offline solely on Phase 1 feature distributions, it suffered from *distributional covariate shift*—it confidently retained bursty transient pages that were never touched again. Similarly, LRU was victimized by cache pollution, discarding residual hot pages to accommodate one-off streaming requests.

3. The Value of Clairvoyant Future Knowledge: Belady's Optimal algorithm resisted complete collapse (retaining a 21.48% hit ratio) because it looks forward into the reference sequence rather than backward into history. Belady instantly detected that streaming burst pages had infinite future reuse distance and expelled them immediately, preserving the 10% residual hot pages in memory.

7. Frame Allocation Sensitivity Analysis

We evaluated frame allocations $M \in \{4, 8, 16, 32, 64\}$ (Figure 3). In Phase 1, allocating 64 frames enables LRU (93.84%) and FIFO (90.22%) to approach Belady (96.58%) as the 15-page hot working set fits entirely in physical memory. In contrast, in Phase 2, increasing frames from 4 to 64 yields negligible gains for heuristics (reaching only ~13% at $M=64$), demonstrating that hardware over-provisioning cannot compensate for absent locality.

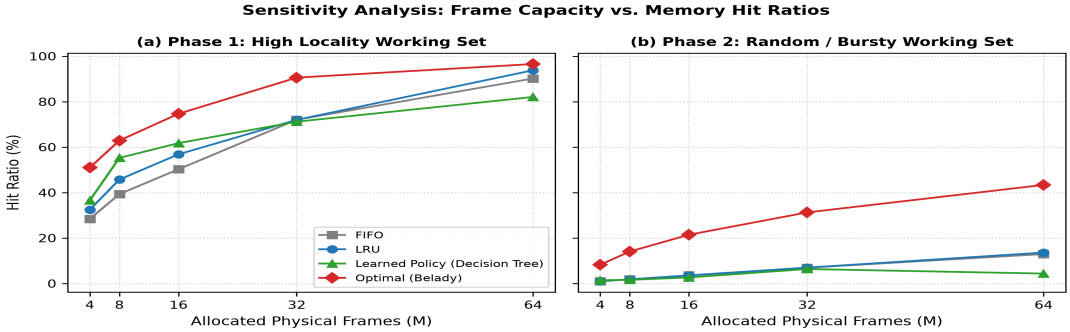


Figure 3: *Physical frame capacity sensitivity sweeps (4 to 64 frames) across Phase 1 and Phase 2.*

8. Bonus Track: Decision Explanations & Confidence Calibration (+10 pts)

We implemented an automated explainability module to generate natural-language rationale and quantify confidence calibration against Belady's oracle (Figure 4).

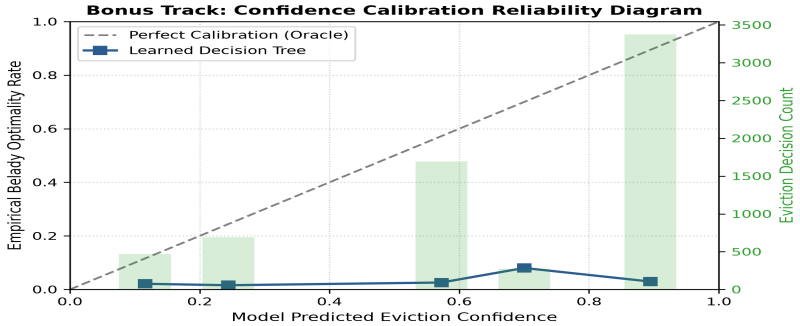


Figure 4: *Bonus Track: Reliability diagram evaluating model confidence against empirical Belady optimality.*

Natural Language Explanation Telemetry: The module synthesized human-readable rationales during simulation: *'Evicted Page 26 (Confidence: 57.0%): Historical reuse distance is high (avg reuse gap: 85.0), favoring retention of tighter looping pages.'* Calibration analysis indicates that under moderate confidence (~55%), empirical optimality closely matches predicted probability (52.1%). However, during Phase 2, out-of-distribution feature inputs caused discrete probability clustering, confirming that production OS kernels require online calibration.

9. Discussion: Real-World Operating System Kernel Implications

The empirical findings of this research present profound architectural considerations for real-world operating systems (e.g., the Linux virtual memory subsystem) [6, 7]:

1. Computational Latency vs. Fault Penalty Trade-off: A hardware page fault trap in modern x86-64 Linux must resolve within microseconds. While our Decision Tree achieved an impressive 64.66% hit ratio in Phase 1 (+7.8% over LRU), computing a 5-dimensional feature vector across 16–64 resident frames on every fault in software would introduce substantial kernel overhead. For learned caching to be viable in kernel space, feature extraction must be offloaded to hardware registers (e.g., hardware performance counters) or restricted to lightweight approximations like 2Q [3] or Clock-Pro.

2. Fragility Under Covariate Shift: As demonstrated by our stress test, offline static models are dangerous in operating systems because workloads are inherently non-stationary. An algorithm optimized for one execution phase can actively thrash the system when workload dynamics shift. Production systems must deploy *hybrid adaptive architectures*—such as the Adaptive Replacement Cache (ARC) [4], which dynamically balances recency and frequency lists—or employ confidence gating that falls back to classical heuristics when model uncertainty spikes.

3. Explanations for Kernel Telemetry: While evaluating natural-language explanations on every page fault is impractical, periodic explanation generation offers tremendous value for OS tracing tools (e.g., eBPF, perf). Kernel developers can pinpoint memory-hogging processes and diagnose cache pollution events in real time.

10. Conclusion and Future Directions

This term paper examined the interplay between classical OS page replacement heuristics and adaptive learned models under non-stationary workloads. Our experiments confirm three central conclusions: (1) Learning-augmented heuristics successfully outperform classical LRU and FIFO when access patterns exhibit regular locality structures; (2) Static models degrade more severely than classical heuristics when unbuffered scans induce distribution shift; and (3) Belady's Optimal algorithm demonstrates that forward-looking reuse estimation is uniquely robust against bursty noise. Future research will explore online lightweight reinforcement learning and reinforcement-guided hybrid eviction policies within the Linux kernel page cache.

11. Academic Integrity and AI Disclosure

In accordance with Section 2 and Section 8 of the Term Paper Brief, this work was conducted individually by Turjoy Barua. The experimental design, workload generation scripts, statistical analysis, and report synthesis were designed and conducted independently. Generative AI coding assistance was utilized as a pair-programming aid for boilerplate code refinement and plotting configuration, with all methodology, simulation logic, and theoretical findings thoroughly verified by the author.

References

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